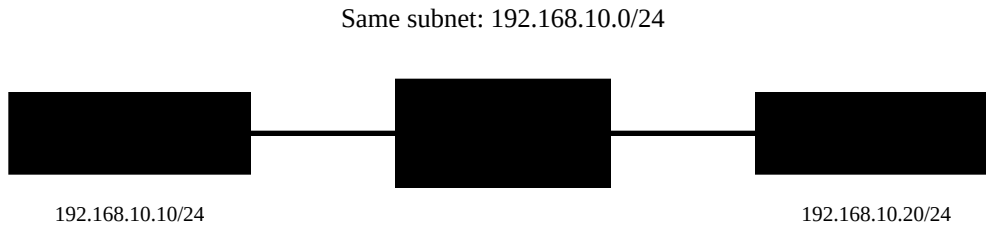


Determining Valid IPv4 Networks

Cisco Packet Tracer Activity – Documentation

Objective: Practice identifying valid IPv4 addresses, subnet masks, and gateways, then verify communication using ping.



Topology: Two PCs connected through one switch. No router is required for communication within the same subnet.

1. Working IPv4 Configuration

Device	IPv4 Address	Subnet Mask	Default Gateway	Expected Result
PC0	192.168.10.10	255.255.255.0	192.168.10.1	Working
PC1	192.168.10.20	255.255.255.0	192.168.10.1	Working

Note: Because both PCs are in the same /24 network, the default gateway is not used for PC0-to-PC1 communication. It may be entered as 192.168.10.1 for the configuration exercise, but there is no router in this topology.

2. Packet Tracer Test – Working Address

On PC0: Desktop → Command Prompt → enter **ping 192.168.10.20**.

Expected result: Replies are received from 192.168.10.20, showing that the two PCs can communicate through the switch.

3. Non-Working Scenario A – Different Network

Device	IPv4 Address	Subnet Mask	Expected
PC0	192.168.10.10	255.255.255.0	Source
PC1	192.168.11.20	255.255.255.0	Ping fails

Why it fails: PC0 and PC1 are in different /24 networks. A switch forwards frames inside the local LAN, but a router is needed to route traffic between different IP networks.

4. Non-Working Scenario B – Wrong Subnet Mask

Device	IPv4 Address	Subnet Mask	Expected
PC0	192.168.10.10	255.255.255.0	Source

PC1	192.168.10.20	255.255.255.128	May fail depending on addresses
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For a clearer guaranteed mask-mismatch test, use PC1 = 192.168.10.130 with mask 255.255.255.128 while PC0 remains 192.168.10.10/24. The hosts can interpret the network differently, so communication can fail because of incorrect subnet logic.

5. Non-Working Scenario C – Invalid/Incorrect Gateway

For the two-PC same-subnet test, changing the gateway to an unrelated address does **not necessarily stop PC0 from pinging PC1**, because the gateway is only needed when traffic leaves the local subnet. This is an important observation for the explanation.

6. What to Capture in Cisco Packet Tracer

- Screenshot 1: Full Packet Tracer topology showing PC0, the switch, PC1, and green connection indicators.
- Screenshot 2: PC0 IP Configuration showing 192.168.10.10 and 255.255.255.0.
- Screenshot 3: PC1 IP Configuration showing 192.168.10.20 and 255.255.255.0.
- Screenshot 4: PC0 Command Prompt showing a successful ping to 192.168.10.20.
- Screenshot 5: A failed test using one of the non-working configurations, together with the reason it failed.

7. Short Conclusion

The valid configuration places both PCs in the same IPv4 network, 192.168.10.0/24, allowing them to communicate through the switch. When the IP addresses or subnet masks are configured incorrectly, the PCs may no longer recognize each other as being on the same local network. The ping command provides a simple way to verify whether the configuration is working.

Important: This document provides the exact configuration and expected test results. The actual Packet Tracer screenshots should be captured from the student's running Packet Tracer file and inserted into the final submission.